

VinBigData Chest X-ray Abnormalities Detection

DATASET DESCRIPTION:

The VinBigData Chest X-ray Abnormalities Detection dataset is a large-scale medical imaging dataset used for detecting thoracic abnormalities in chest X-ray images. The dataset contains high-resolution chest X-rays along with bounding box annotations for multiple abnormality classes. The dataset is designed for object detection tasks in the medical imaging domain.

Annotation Columns:

- `image_id` – Unique identifier of the image
- `class_name` – Name of abnormality
- `class_id` – Numerical label of abnormality
- `x_min`, `y_min` – Top-left corner of bounding box
- `x_max`, `y_max` – Bottom-right corner of bounding box

Image Data

- Chest X-ray images (originally in DICOM format)
- Grayscale medical images

Numerical Data

- Bounding box coordinates
- Class IDs

Textual Data

- Image identifiers
- Disease class names

DATA PREPROCESSING:

2.1 Introduction to Data Preprocessing

Data preprocessing is a crucial step in deep learning pipelines. Medical images require special preprocessing to ensure:

- Consistent input dimensions
- Enhanced contrast for abnormality visibility

- Reduced noise
- Proper annotation scaling

Preprocessing ensures that the data becomes machine-friendly while preserving diagnostic information.

DICOM to JPG Conversion

Chest X-ray images were originally stored in DICOM format. These were converted to JPG format

```
input_folder = "/kaggle/input/vinbigdata-chest-xray-abnormalities-detection/train"
```

```
for image_id in tqdm(selected_images):
    dicom_path = os.path.join(input_folder, image_id + ".dicom")
    # Read DICOM file
    dicom = pydicom.dcmread(dicom_path)
    image = dicom.pixel_array
    # Convert to 8-bit format (required for JPG)
    image = image.astype(np.float32)
    image = (image - image.min()) / (image.max() - image.min())
    image = (image * 255).astype(np.uint8)
    # Save as JPG
    output_path = os.path.join("train_jpg", image_id + ".jpg")
    cv2.imwrite(output_path, image)
    print("Conversion completed successfully")
```

Grayscale Conversion: Chest X-ray images are naturally grayscale. If any image is in color format, it is converted to grayscale. This reduces unnecessary information and ensures the model focuses only on intensity variations in the lungs.

Image Resizing: All images are resized to a fixed size (512×512 pixels).

Intensity Normalization: Pixel values originally range from 0 to 255. These values are scaled between 0 and 1.

Contrast Enhancement (CLAHE):CLAHE (Contrast Limited Adaptive Histogram Equalization) is applied to improve image contrast.

Noise Reduction:A light denoising technique (Gaussian filtering) is applied. This removes small unwanted noise while keeping important medical details intact.

Conversion to YOLO Format:Bounding box annotations are converted into YOLO format.

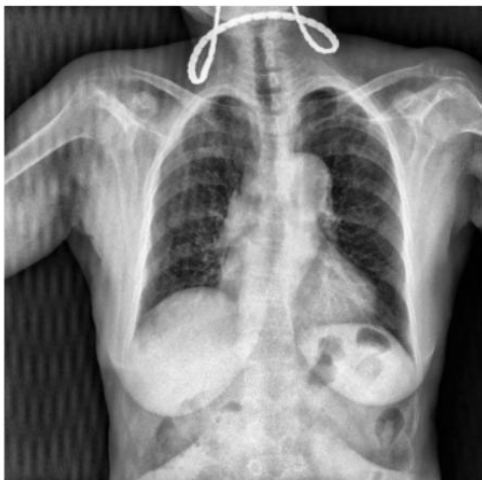
YOLO format represents bounding boxes using:

- Class ID
- Center coordinates (x, y)
- Width and height

All values are normalized between 0 and 1.



CLAHE Enhanced



Augmented

